

COURSE TITLE: MEASUREMENT AND EVALUATION

LEVEL : 4

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COURSE OBJECTIVES

At the end of this course, learners should be able to know what measurement, assessment and evaluation are, and know the importance of an evaluation. They should equally understand principles and concepts of different types of pedagogic evaluations.

They should moreover be able to elaborate good pedagogic objectives, to understand the relationship between pedagogic objectives and evaluation.

Learners should also master the process of an evaluation. They should be able to plan and administer evaluations, taking into consideration established principles and effective practices, and know how to make use of the results obtained in evaluations.

Finally, learners should be aware of the biases of an evaluation

Definitions

Measurement

Assignment of numbers (quantity). It is the process of obtaining numerical description of the degree of what an individual possesses.

Drucker« If you can't measure it, you can't improve it »

- - "Measurements allow a more objective description of character traits and facilitate comparisons. “ Osman
 - Can we measure / quantify everything? ex motivation
Yes even if it is difficult "if something exists, it exists in quantity; if it exists in quantity, it can be measured. “ scholars
 - However, it must be recognized that it is difficult to have valid measurements (which is used to measure). We base ourselves on deduction.

ASSESSMENT

Process by which information is obtained relative to objective. It is the process of gathering and organizing data (the basis for decision making)

EVALUATION

It is the process of obtaining, analyzing and interpreting information to determine the extent to which students achieve instructional objectives.

According to Osman, it is collection and systematic analysis of data, so as to take educational decisions.

Note that decisions are taken on the basis of some criteria (certain standard on which the achievement of a learner is measured)

TEST

A test is a fast way used to measure the skill, knowledge, intelligence, capacities or aptitudes of an individual or group.

FUNCTIONS OF EVALUATION

Evaluation assesses or makes appraisal of:

- educational objectives, instructional materials, programs, curricula, facilities
- teacher
- learner
- school institution
- educative policy

It also gives information to the educative community.

Functions of measurement

It helps to determine

- what knowledge, skills, abilities, habits and attitudes have been acquired
- what progress or extent of learning attained
- the strengths, weaknesses, difficulties and needs of students.

PRINCIPLES OF EVALUATION

An evaluation should be

- based on clearly stated objectives
- comprehensive
- used judiciously
- continuous and integral part of the teaching-learning process

Types of evaluations

Non intrusive and intrusive

Musial and al. (2012)

Intrusive: This interruption and the evaluation should be beneficial for learners. Waste of time, stressful for the learner.

Non intrusive: during a lesson, a week, a month or even a year. No interruption in learning. More time to learn than to being evaluated, and for the teacher more time to plan and teach than to conceive evaluations and marking copies. Appropriate for very young learners. **Eg of cases:** level of interaction, noise making in class, hygiene in the market place, etc.

Advantages and disadvantages

Diagnostic, formative and summative

Time for evaluation

Diagnostic: It is usually given before instructions/teaching

It identifies the weakness and strength of an individual.

Formative: ongoing evaluation during an instructional period. Used to determine the effectiveness of teaching and learning (if they are progressing towards the stated objectives...). Benchmarking for professional growth (Marczely, 2001).

Summative: Conducted at the end of course. Measures the extent to which students have attained the fixed objectives. Used for making administrative decisions (promotion, educational guidance, failure and even dismissal)

Placement evaluation

It determines students' entry behaviour. It defines knowledge and skills he possesses which are necessary to be able to follow up a programme. (entrance exams: ENS, police, medical schools, etc.)

To determine the grade or year level the pupil or student should be enrolled.

Objective and subjective

Objective evaluation: This is a test with definite and clear answers. The learner does not need to construct answers, but to select the correct answers. Eg: True or false, multiple choice questions.

Adv and disadvantages

Subjective : The score is affected by bias or subjectivity. The most common form is essay.

Adv and disadvantages

Written, practical and oral (Kusum)

Written: the learners or the respondents are requested to provide a written answer eg: any written evaluation, a questionnaire

Practical: the learners or the respondent are requested to carry out a practical activity. Eg: project assignment, case study, observation, interview, Focus group discussion (FGD)

Oral: Answers should be oral; Eg: interview, FGD

Standardized and teacher made tests

Standardized test: Same test given in the same manner to all test takers, and graded in the same manner for everyone. They are made by experts in the field concerned and in the field of measurement and evaluation.

Eg: Intelligence quotient, personality test.

Teacher made test: constructed by a teacher

Norm reference test and criterion-reference test

Norm reference: refers to a method of evaluation that produces results allowing an educator to see if a student or group of students have performed better or worse than a controlled, reference group. Eg: baby weight growth charts, IQ (discussion) min=70, average=90-109, geniuse= more than 140

Advantages and disadvantages

Criterion-reference test: they compare a student's knowledge and skills against a predetermined standard, cut score or other criterion. The performance of other students does not affect a student's score.

In conclusion

In case you decide to make an intrusive evaluation, you should establish a clear contract with your learners, telling them before the evaluation:

- What is going to be evaluated ?
- how (which type of activities, type of evaluation)?
- What will the criteria be?

Learning objectives

DEFINITION

Saul and al.

From an educational perspective, learning objectives are statements of intent that describe what a student will be able to do as a result of learning that he could not do before.

Rai and Osman

Educational objectives are the statements of those desired changes in behavior as a result of specific teaching-learning activity of teacher-learner activities.

IMPORTANCE OF LEARNING OBJECTIVES

If you don't know where you are going to, you might end up where you don't know.

- n Provide a framework for organizing teaching.
- n Help teachers focus on “need to know” content and eliminate unnecessary content.
- n Inform learners about what new information or skills they will be learning. Learners can take more responsibility for their own learning when they know what they are expected to achieve, and better prepare for evaluation and carry on metacognitive activities (self eval)
- n Guide teachers in choosing appropriate methods and materials to enhance learning.
- n Assist teachers in designing appropriate assessment and evaluation tools that accurately reflect the curriculum, and inform whether or not outcomes have been achieved.
- n Support meaningful review of curricula, leading to more balanced and well-sequenced teaching products.
- n Ensure that the desired program goals and objectives are achieved.

Different types of LO

The terminology may differ from one scholar to another.

According to Osman, there are:

- general objectives / goals (Saul et al.) // central objectives
- intermediate objectives / objectives // contributory objectives
- specific objectives / outcomes // indirect objectives/// operational

General objectives

They are general statements concerning the overall goals, ends or intentions of learning.

These are very general statements which describe in implicit terms the overall aims of the total educational process. * **note the verbs used**

Set at the course or program level to inform curricular design, sequencing and assessment

Examples: On completion of the course/study/programme, the student should be able to:

- understand the concept of electromagnetism
- to appreciate the difference between individual and group learning
- to comprehend the use of a scanner

Intermediate objective

They are more specific than g.o. They describe specific steps that need to be taken to meet the overall goals of the course or the program.

Made by breaking down general objectives into components.

They are set at the beginning of each sequence of the course (chapters, parts, unit, module)

Eg: At the end of this chapter, the student should be able to know how a successful group learning is organised.

Specific objectives

They specify what learners should be able to do by the end of a session or a lesson. They are more specific than intermediate objectives. They are more precise and more student centered.

Eg: By the end of this lesson, students should be able to list five factors to take into consideration when organising a learning group.

Domains of learning objectives

Benjamin Bloom's taxonomy of educational objectives (1956): System of classification of the various domains of human learning

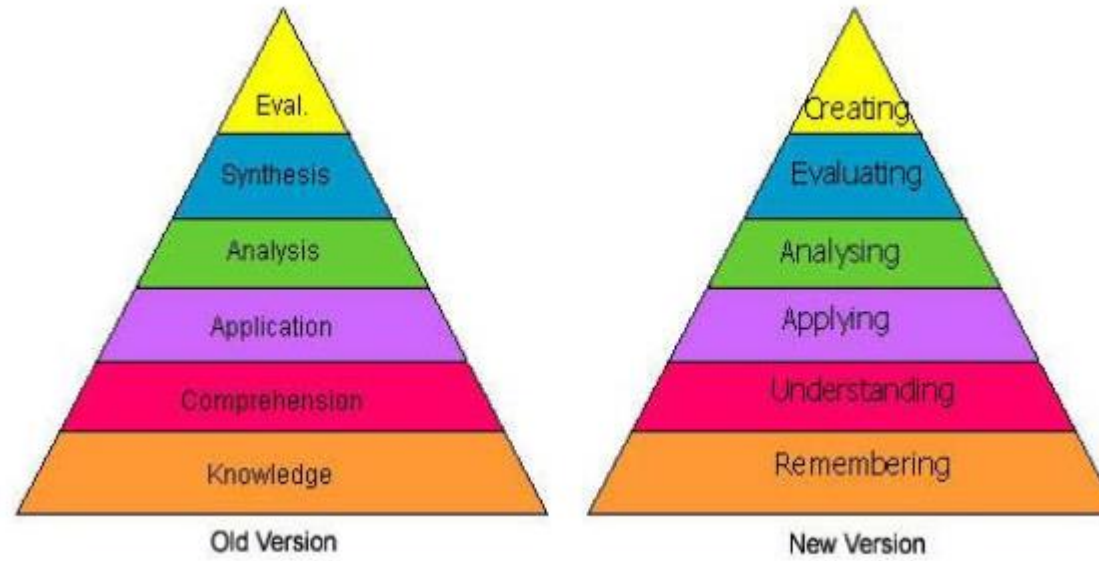
Learning and mastery of a subject require more than simply acquiring new knowledge or facts.

Educators often classify learning into three domains or categories based on a classification system created by Benjamin Bloom and associates at the University of Chicago.

These include cognitive (knowledge), affective (attitude), and psychomotor (skill); each domain includes subcategories that move from simple to more complex processes.

The cognitive domain

The cognitive domain relates to knowledge and intellectual skills : from simple cognitive processes to complex ones.



The affective domain

The affective domain relates to the emotional component of learning, and is concerned with changes or growth in interest, attitudes, and values

- . It emphasizes feeling, tone, emotion, or degree of acceptance or rejection.

Subcategories classified from more simple affective components to more complex: receiving (willing to listen), responding (willing to participate), Valuing (willing to be involved), Organization (willing to be an advocate), Internalizing values (willing to change one's behavior)

The psychomotor domain

The psychomotor domain relates to the physical skills and/or the performance of motor tasks according to a standard of accuracy, rapidity, or smoothness.

Learning is demonstrated by the learner performing the skill to a designated standard or level of proficiency.

Subcomponents: perception, set, guided response, mechanism, complex overt response

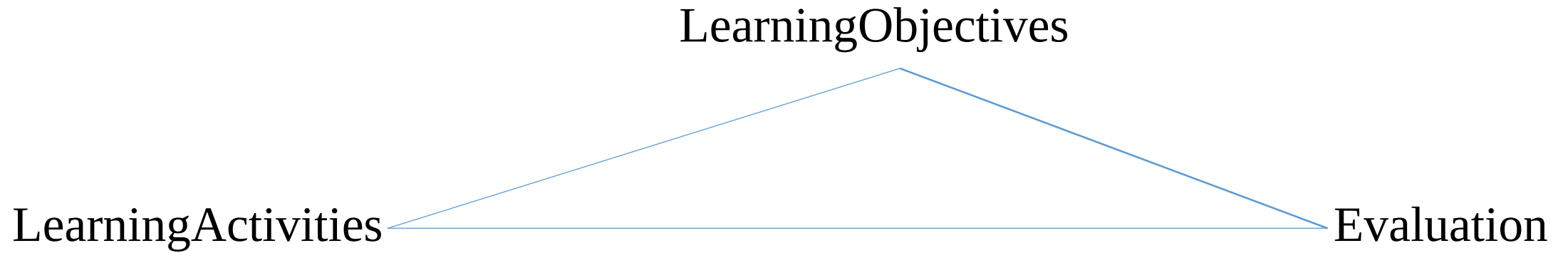
Important !!!

(When designing a program, it is important to think about these three domains of learning and their subcategories, and to develop objectives for the training that address each domain. This will lead to activities that engage the different domains of learning, and will result in a more holistic and comprehensive training.)

It is important to vary the domains and the subcategories. Take into consideration the level of the students and their differences (inclusive classes mostly! = differentiated pedagogy Meirieu)

The magic triangle

It represents the relationship between learning objectives, learning activities and evaluation.



Learning activities are all activities that are developed to foster learning towards reaching the LOs. Evaluation or tests allow the author to check the degree to which the students are reaching the objectives. If these three components are congruent then learning is in an optimal way.

Components of educational specific objectives

Lo should be constructed around 4 main components (Robert Mager):

1-audience: specify the intended audience. Recipients of the lesson: learners

2- behavior: stated in observable and measurable (quantifiable) action verb (so avoid verbs like: understand, know, imagine)

3. Standard or criterion: Determine to what standards the task must be performed. Standards describe the level of proficiency to be elicited by the training—in other words, how the learner will be able to perform in terms of quantity, quality, and time limitations

4- condition: Determine under what conditions the task must be performed. The actual conditions under which the task will occur or be observed speak to time and place. The condition might also identify tools, procedures, materials, aids, or facilities to be used in performing the task, for example, “without reference to a manual” or “by checking a chart.”

LO should be SMART

Specific: precisely states what the learner will be able to do

Measurable: can be observed or counted during or after the training session

Action-oriented: uses an active verb that represents a behavior change or acquisition

Reasonable: is appropriate to the time and scope

Time-bound: can be achieved by the end of the teaching

Example of a good specific objective

At the end of this lesson, the learner should be able, in three relevant points, to explain the relationship between poverty and school failure.

Audience: the learner

Behaviour: explain

Standard or criterion: three, relevant

Condition: At the end of this lesson

Practice

Choose any topic of your field of study:

construct one general objective, from it, construct one intermediate objective, from it, construct 3 good specific objectives

QUALITIES OF A GOOD MEASURING INSTRUMENT

Regardless of the type of tool used or how the results of evaluation are to be used, all types of evaluation should possess certain characteristics. The most important characteristics are:

1-Validity: It refers to the extent to which a test measures what it is supposed to measure. Although there are several kinds of validity, the two types commonly used are criterion and content.

***criterion validity** (or criterion-related validity) measures how well one measure predicts an outcome for another measure. A test has this type of validity if it is useful for predicting performance or behavior in another situation (past, present, or future).

Content Validity is a process of determining the extent to which a set of test tasks represent a domain of tasks about which interpretations of test scores are made. The essence of content validation is determining the adequacies of the sampling of the content that the test scores are interpreted to represent.

2-Reliability: It is concerned with the extent to which an evaluation test is consistent in measuring what it is intended to measure. Whether the response of the respondents would have been different if they were tested yesterday or next week? How would the result have varied if a different sample of equivalent items had been selected? If the measurements are not consistent over different occasions or over different samples of the same performance domain, the evaluator can have little confidence in the results.

Some methods of estimating reliability:

Test-Retest Method - In this test, the same tool/ instrument is administered to the same sample on two different occasions. The resulting test scores are correlated and the correlation coefficient provides a measure of stability over a given period of time. If the results are highly stable, those respondents who are high on one administration of test will also be high on the other administration and the other respondents tend to remain in their same relative positions on both administrations. An important factor to be kept in mind is the time interval between tests when interpreting measures of stability. If the time interval is short (say 1-2 days), the consistency of results will be inflated because respondents will remember some of their answers from the first test. If the time interval is quite long (say 1 year), the results will be influenced by the actual changes in the respondent over that period of time. Therefore, the best time interval between test administration will mainly depend on the use to be made of results.

Equivalent-Forms Method - This method uses two versions of an instrument given to the same sample of respondents. The two forms of the instrument are administered to the same group of respondents in close succession, and the resulting scores are correlated. The correlated coefficient provides a measure of equivalence. It indicates the degree to which both forms of the test are measuring the same aspects of behaviour. The equivalent forms method reflects short term constancy of respondents' performance and the extent to which the test represents an adequate sample of the characteristics being measured.

To optimize reliability, make sure that:

- one evaluative question tests only one thing at a time
- questions adequately reflect the objectives of the teaching unit.

3-Practicability: A good evaluation should be realistic, practical and efficient in terms of their cost, time taken and conditions of application (make sure that **the conditions of evaluation are conducive**)

4-Fairness: Evaluation must be fair to all students. To keep fairness in evaluation, it is also desired that students should know exactly how they are to be evaluated. This means that students should be provided information about the evaluation such as the nature of the materials on which they are to be examined (i.e., Context and Objectives), the form and structure of the examination, length of the examination and the value (in terms of marks) of each component of the course.

5-Usefulness: Evaluation should also be useful for students. Feedback from evaluation must be made available to the students and weakness. By knowing their strength and weakness, Students can think of further improvement. Evaluation should suggest all the needful requirements for their improvement.

6-Interpretation of Results: Another factor which must be considered in the choice of a test is the ease of interpretation of test results. A test score is not meaningful unless the teacher or counselor is able to decide what significance or importance should be attached to it and to make some judgment concerning its relationship to other kind of information about the student.

TABLE OF SPECIFICATIONS (TOS)

Definition

A Table of Specifications is a plan prepared by a classroom teacher as a basis for test construction especially a periodic test. It is a two-way chart which describes the topics to be covered by a test and the number of items or points which will be associated with each topic. It is a tool that allows the teacher to construct a test which focuses on the key areas and weights those different areas based on their importance. A Table of Specifications provides the teacher with evidence that a test has content validity, that it covers what should be covered.

A sample table of specifications for Fifth Grade social studies, chapter 6: The middle colonies

A	B	C	D	E	F	G
	Instructional Objectives	Time Spent on Topic (minutes)	Percent of Class Time on Topic	Number of Test Items: 10	Lower Levels -Knowledge -Recall -Identification -Comprehension	Higher Levels -Application -Analysis -Evaluation -Synthesis
Day 1	1. Identify the various groups who settled the Middle Atlantic Colonies.	15	10.00%	1.00	1 Multiple Choice	
	2. Summarize the contributions of different religious and cultural groups to the settlement of the Middle Atlantic Colonies.	15	10.00%	1.00	1 Short Answer	
Day 2	3. Identify George Whitefield as an early leader of the Great Awakening	10	6.70%	.67	1 Multiple Choice	
	4. Evaluate the impact of the Great Awakening sermons on English colonists.	20	13.30%	1.33		1 Multiple Choice
Day 3	5. Describe the physical features that helped Philadelphia become a main port.	15	10.0%	1.00	1 Multiple Choice	
	6. List ways in which immigrants aided Philadelphia's growth and prosperity.	10	6.70%	.67	1 Short Answer	
	7. Identify the contributions Benjamin Franklin made to Philadelphia.	5	3.30%	.33	---	
Day 4	8. Interpret information in a circle graph.	15	10.00%	1.00		1 Multiple Choice
	9. Gather and organize information using a circle graph.	15	10.00%	1.00		1 Short Answer
Day 5	10. Identify the challenges faced by backcountry settlers.	5	3.30%	.33	---	
	11. Analyze the importance of the Great Wagon Road as an early transportation route.	10	6.70%	.67		1 Multiple Choice
	12. Explain how backcountry settlers adapted to and made use of the resources available to them.	15	10.00%	1.00		1 Multiple Choice

How to create a table of specifications?

From what we have said so far, it is evident that the table of specification helps you plan out your evaluation. It is made in five easy steps:

-Step 1 Determine the coverage of your exam.

make sure the coverage of your exam is something that you have satisfactorily taught in class. Select the topics that you wish to test in the exam. It is possible that you will not be able to cover all these topics as it might create a test that is too long and will not be realistic for your students in the given time. So select only the most important topics.

-Step 2 Determine your testing objectives for each topic area.

So for each content area that you wish to test, you will have to determine how you will test each area. Will you test simply their recall of knowledge? Or will you be testing their comprehension of the matter? Or perhaps you will be challenging them to analyze and compare and contrast something. Again, this would depend on your instructional objectives in the classroom. Did you teach them lower thinking skills or did you challenge them by making them think critically?

-Step 3 Determine the duration for each content area.

Write down how long you spent teaching a particular topic. This is important because it will determine how many points you should devote for each topic. Logically, the longer time you spent on teaching a material, then the more questions should be devoted for that area.

-Step 4 Determine the Test Types for each objective.

Determine the test types that will accomplish your testing objectives. For example, knowledge questions can be accomplished easily through multiple choice questions or matching type exams. If you want to test evaluation or synthesis of a topic, you will want to create exam type questions or perhaps you will ask the students to create diagrams and explain their diagrams in their analysis. The important thing is that the test type should reflect your testing objective.

-Step 5 Polish your terms of specification.

After your initial draft of the **table of specifications**, it's time to polish it. Make sure that you have covered in your terms of specification the important topics that you wish to test. The number of items for your test should be sufficient for the time allotted for the test. Then it's time to put into action your blueprint by creating your exam.

WRITING EVALUATION ITEMS

As already mentioned, evaluation is an important aspect of the teacher's activities. Therefore, it should be well planned and conducted perfectly. Planning an evaluation includes formulating good and effective questions. This requires a certain expertise in the matter.

I-Writing objective evaluation questions.

1- Multiple choice questions

A multiple choice item consists of a problem, known as the stem, and a list of suggested solutions, known as alternatives. The alternatives consist of one correct or best alternative, which is the answer, and incorrect or inferior alternatives, known as distractors.

How to construct a good stem?

a-The stem should be meaningful by itself and should present a definite problem. A stem that presents a definite problem allows a focus on the learning outcome. A stem that does not present a clear problem, however, may test students' ability to draw inferences from vague descriptions rather than serving as a more direct test of students' achievement of the learning outcome.

b-The stem should not contain irrelevant material, which can decrease the reliability and the validity of the test scores.

c-The stem should be negatively stated only when significant learning outcomes require it. Students often have difficulty understanding items with negative phrasing. If a significant learning outcome requires negative phrasing, the negative element should be emphasized with italics or capitalization.

d-The stem should be a question or a partial sentence. A question stem is preferable because it allows the student to focus on answering the question rather than holding the partial sentence in working memory and sequentially completing it with each alternative. The cognitive load is increased when the stem is constructed with an initial or interior blank, so this construction should be avoided.

How to construct effective alternatives?

a-All alternatives should be plausible. The function of the incorrect alternatives is to serve as distractors, which should be selected by students who did not achieve the learning outcome but ignored by students who did achieve the learning outcome. Alternatives that are implausible don't serve as functional distractors and thus should not be used. Common student errors provide the best source of distractors.

b-Alternatives should be stated clearly and concisely. Items that are excessively wordy assess students' reading ability rather than their attainment of the learning objective.

c-Alternatives should be mutually exclusive. Alternatives with overlapping content may be considered "trick" items by test-takers, excessive use of which can erode trust and respect for the testing process.

d-Alternatives should be homogenous in content. Alternatives that are heterogeneous in content can provide cues to student about the correct answer.

e-Alternatives should be free from clues about which response is correct. Sophisticated test-takers are alert to inadvertent clues to the correct answer, such as differences in grammar, length, formatting, and language choice in the alternatives. It's therefore important that alternatives

- have grammar consistent with the stem.
- are parallel in form.
- are similar in length.
- use similar language (e.g., all unlike textbook language *or* all like textbook language).

f-The alternatives should be presented in a logical order (e.g., alphabetical or numerical) to avoid a bias toward certain positions.

g-Keep the specific content of items independent of one another. Savy test-takers can use information in one question to answer another question, reducing the validity of the test.

2- True or false items

Here, a factual statement is made and the learner is required to respond with either true or false depending on the correctness of the statement. The following are some guidelines (Worthen et al., 1999) to construct good true/false questions:

a- **Avoid double-barreled questions. Write questions that ask about a single idea.**

Eg:

a-1-The true-false item is also called an alternative-response item.

- True
- False

a-2-The true-false item, which is favored by all test experts, is also called an alternative-response item.

- True
- False

Answer

- The question contains two ideas and therefore is confusing.

b-Avoid using negative wording in false statements

b-1. Tuberculosis is a communicable disease.

- True
- False

b-2. Tuberculosis is not a noncommunicable disease.

- True
- False

Answer

- B-1 is better. 2B is confusing because of the double negative.

c-Use approximately the same number of true and false questions

d-Arrange the true and false questions in a random order

e-Keep the lengths of the true and false questions about the same

f-Avoid writing long and complicated questions. Break them into simple sentences.

f-1. The true/false item is more subject to guessing but it should be used in place of a multiple-choice item, if well constructed, when there is a dearth of distracters that are plausible.

- True
- False

f-2. The true/false item should be used in place of a multiple-choice item when only two alternatives are possible.

- True
- False

Answer

f-2 is better. The language in f-2 is simple, clear, and concise.

g-Avoid using absolute terms or superlatives such as always, never, largest, best, usually, generally, etc.

g-1. A statement of opinion should never be used in a true/false item.

- True
- False

g-2. A statement of opinion, by itself, cannot be marked true or false.

- True
- False

Answer

g-2 is better. True/false statements should avoid absolutes.

h-Do not include irrelevant material.

3-Matching items

The matching test item format provides a way for learners to connect a word, sentence or phrase in one column to a corresponding word, sentence or phrase in a second column. The items in the first column are called *premises* and the answers in the second column are the *responses*. The convention is for learners to match the premise on the left with a given response on the right. By convention, the items in Column A are numbered and the items in Column B are labeled with capital letters.

Eg:

<i>Premises</i> Column A	<i>Responses</i> Column B
___1. Person who performs mysterious tasks no one understands	A. Facilitator
___2. Person who provides schooling for children	B. Trainer
___3. Person who enables a group to find solutions	C. Instructional Designer
___4. Person who instructs adults in a classroom.	D. Meeting Organizer
	E. Teacher

If you decide to use a matching format, take the time to construct items that are valid and reliable. Here are some guidelines for this.

a-Two-part directions. Your clear directions at the start of each question need two parts: 1) how to make the match and 2) the basis for matching the response with the premise. You can also include whether items can be re-used, but often pre-built templates don't allow for this.

Example above: *Drag each career name in Column B to the best definition in Column A. No items may be used more than once.*

b-Parallel content. Within one matching test item, use a common approach, such as all terms and definitions or all principles and the scenarios to which they apply.

c-Plausible answers. All responses in Column B should be plausible answers to the premises in Column A. Otherwise, the test loses some of its reliability because some answers will be “give-aways.”

d-Clueless. Ensure your premises don't include hints through grammar (like implying the answer must be plural) or hints from word choice (like using the term itself in a definition).

e-Limited premises. Due to the capacity limitations of working memory, avoid a long list of premises in the first column. A number that I've come across is to keep the list down to six items. Even less might be better, depending on the characteristics of your audience.

f-One correct answer. Every premise should have only one correct response. Obvious, but triple-check to make sure each response can only work for one premise.

4-completion items and short answer test items or closed questions

Completion items require a word, short phrase, number, or symbol response. They are sometimes called *fill-in-the-blank questions*. Sometimes a word bank accompanies a set of short-answer exercises, especially for younger children or where recognition (not production) of the word is important and spelling is not. Use completion items especially when recall of a specific word or term is important for the learning goal. Write effective completion items by following the guidelines below:

a- Write each item to tap into the content and kind of thinking in important learning outcomes. Make sure the alignment between the item and the learning outcome is strong.

b-Design items so that students must supply an important word or concept. Write in a question format if possible. Direct questions are more likely to have unambiguous answers than sentences with blanks in them. Design the items so that the particular important word or phrase that is required would be clear to students who know the material.

c-Insert a blank where an answer is expected. If the item is a direct question, a blank should be placed after the question. If the item is a sentence that leaves out a key word or idea, a blank that serves as a placeholder for that item should be included and it should come toward the end of the sentence. When the missing element is placed near the end of the sentence, students can read through the sentence once and gather enough context to answer without having to backtrack and read the sentence again.

d-Avoid grammatical or other clues. Clues from grammar or from the length of the line confound other kinds of reasoning with the knowledge or skill the item is intended to assess. In a set of completion items, keep all blanks the same length so that students don't infer that some answers are long and others are short. In addition, avoid grammar clues within the item. For example, the item "A person who draws the pictures for a book is called an _____." suggests that the correct answer begins with a vowel sound.

Short answer questions have a slight advantage In the "Remember" category of Bloom's taxonomy. Here are some guidelines for writing good short answer questions:

- a. Word the question so that a clear, meaningful problem is presented.
- b. Structure the problem so that the range of acceptable responses is limited to a single correct answer or a narrow set of definite, clear-cut, and explicit answers. If you don't do this, be prepared to accept any answer that could be correct in your broadly defined situation.
- c. Generally, use direct questions rather than incomplete sentences.
- d. Focus on eliciting key words or ideas only. Do not use these questions for random peripheral vocabulary or ideas. Also, don't use short answer when you really want an essay.
- e. Avoid using statements taken directly from textbooks, reference manuals, or other documents with one or two words omitted (swiss cheese). Use paraphrased vocabulary and grammatical structures that are independent of the context and textbook expressions of the source document.
- f. Avoid using excessive blanks in any single item.
- g. Avoid providing extraneous clues - see the section on multiple choice for examples.

II-Writing subjective evaluation questions.

In this kind of test the objective is to measure qualities such as learner's ability to perform certain practical or intellectual skills which might include describing something accurately either in oral or written form using materials imaginatively, working creatively, handling information logically building convincing arguments or exposing flaws in the arguments of others. The common types of subjective test items that we have are:

1-Restricted response items

On restricted response items examinees provide brief answers, usually no more than a few words or sentences, to fairly structured questions.

2-Extended response items

Here items require lengthy responses that count heavily in scoring. These items focus on major concepts of the content unit and demand higher level thinking. Examinees must organize multiple ideas and provide supporting information for major points in crafting responses.

The construction of the above types of tests items has a detailed process which includes the following stages:

- Developing the prompt
- Creating the scoring rubric
- Scoring response

a-Developing the prompt: The prompt for a subjective item poses a question, presents a problem, or prescribes a task. It sets forth a set of circumstances to provide a common context for framing the response.

Action verbs direct the examinee to focus on the desired behavior, for instance, solve, interpret, compare and contrast, discuss or explain. Appropriate directions indicate expected length format of the response, allowable resources or equipment's, time limits and features of the response that count in scoring.

b-Creating the scoring rubric: These are analytic or holistic in nature.

For analytic rubric, the item writer/ constructor lists desired features of the response with a number of points awarded for each specific feature.

A holistic rubric provides a scale for assigning points to the response based on overall impression.

A range of possible points is specified and verbal descriptors are developed to characterize a response located at each possible point on the scale.

Illustrative responses that correspond to each scale point are often developed or selected from actual examinee responses.

c- Scoring response: During subjective scoring at least four types of rater errors may occur as the rater becomes more lenient or severe over time or scores erratically due to fatigue or distractions; has knowledge or belief about an examinee that influences perception of response; is influenced by examinees good or poor performance on items previously or influenced by the strength or weakness of a preceding examinees response.

THE PROCESS OF AN EVALUATION

I-Planning

It is generally recommended not to design an evaluation in a hurry. The teacher or person responsible for the assessment should have enough time to plan, develop and revise the items.

1-Determine the purpose of the evaluation.

Is it a diagnostic, a formative or a summative evaluation?

2-Determine the nature of the evaluation

Is it going to be an oral, written or practical evaluation? Consider your needs, the nature of the objectives to be evaluated and your situation.

3- Determine the scope of the evaluation

Define what will be covered. Typically, objectives provide the blueprint for the subject or group of subjects that will be included in a test, and a test should be a meaningful unit of instruction. Normally, a test covers the equivalent of several days or weeks of instruction.

Limiting the scope of the test prevents students from experiencing cognitive overload and allows for a much more representative sampling of the behaviors represented by the chosen objectives. Of course, some cumulative tests like final exams require a greater scope and can represent months of instruction. If the individual tests or formative tests have been rigorously designed, the task of developing the cumulative tests is made easier, since the results evaluated by such tests represent a sampling of the returns evaluated by all the other previous tests.

So, at this point, it is necessary to determine the learning objectives that will be evaluated.

4-Prepare the TOS (Table of specifications)

5- Select the appropriate test technique

Chose the format that suits more your needs and your situation. you'll need to base your decision on objective facts and sound principles of assessment. You need to think about the assessment technique that meets the purpose of the test and the objectives to be assessed. Is it going to be an objective or a subjective evaluation? MCQ? True or false? ... restricted response items? Extended response items? A combination of different techniques?

Eg. Take into consideration the size of your classroom, the time allotted for the evaluation, the level of the objectives to assess (low level? high level?), the lenght of time that will be probably spent for the whole activity (subjective evaluation in crowded classrooms for example), etc.

6- Writing the test

Write the different questions/items of the test, bearing in mind the guidelines for any assessment technique chosen. Also remember the characteristics of a good evaluation.

General guidelines for the organization, instructions and format of tests

Organization of test items

- * If there is one type of item, items of the same type should be grouped together.
- * Items can also be organized on a continuum, from least difficult to most difficult.
- Another logical way to organize items is to classify them by subject. All items related to the same objective or outcome should be grouped together.

General Instructions

- * If the purpose and importance of the test is not clear, they should be discussed with the students so that they are guided and motivated to excel.
- * In addition to the instructions for choosing (or providing) the answers, students should be given instructions for recording the answers.
- * If the instructions are complicated or involve a procedure that the students do not know, an example of an item should be presented to the students.

- * A good idea would be for the person designing the test to develop the answer key
- * If certain subjects matter more than others, it is better to have more items for these subjects than to assign a higher value to certain items.
- * If the time required for the test is insufficient, students should be given instructions on how to use it effectively.
- * If there is no penalty for guessed answers, students should be encouraged to answer each item, even if they are unsure of the correct answer.
- * If there is a penalty for guessed answers, students should be instructed to answer only those items for which they are certain they have the correct answer.

Format details

*Items should be numbered consecutively and instructions should be prominently displayed and isolated from the items.

*The spaces for the answers should be the same size and large enough to contain the longest answer, especially for short answer tests, and the spaces for the answers should be placed in a vertical column in the right margin or on the left or on a separate answer sheet.

*If an item is accompanied by any kind of illustration, such as a diagram, it should be right and placed near the item, directly above it if possible or parallel to the item.

Test review

In fact, the review of the test can take place before its intended use as well as after its formal administration. Instruments designed for precise and formal assessment can be administered and revised over and over until they reach acceptable levels of validity and reliability. The process is called "field test" or "test validation".

NB: A good test must be accompanied by instructions explaining how to score and interpret the test results, because other people may take part in the test scoring and interpretation process, due to unforeseen circumstances.

7- Edit your evaluation items

II- Administering the evaluation

1. The conditions in which the evaluation is administered

In general, when administering a test, the following points should be kept in mind:

- *physical disposition: lighting, ventilation, silence, etc.
- *psychological state: avoid causing anxiety related to the test, try to reduce anxiety related to the test, do not give the test when other events are a source of distraction.
- *the suggestions: do not speak unnecessarily before the test, minimize interruptions, do not provide clues to people requesting information on items, discourage cheating, give students the same time to complete the test.

2. Marking student's work: some guidelines

- *Familiarise yourself with the assignment task and the marking guide in the study guide. This is what students have used to guide their work.

*Determine how long you can spend on each assignment and work out how much feedback you intend to give. Long essays will require more comments than a short test or lab report. Remember you are trying to give the students feedback that they can learn from

*Quickly skim the assignment before you start to mark. This gives a sense of the quality of the students work before you start marking.

*Give some early feedback to the whole class via Stream (if appropriate). Give students your initial impressions of the quality of their work after marking a small number of submissions. For assignments and tests where a model answer is available, consider sharing it at this stage. This gives students feedback soon after submission while the assignment is still fresh in their minds.

*If you encounter recurring issues or themes across the class make a note of them and write general feedback for all students. If you know this general feedback is going to all students then you don't need to keep repeating it on every assignment. This will free you to concentrate on areas where students will benefit most from comments on their work.

*Write feedback in a way that is helpful to students. Try to provide constructive comments that help students to identify their weaknesses and give them strategies or ideas about how to improve for next time. When writing your comments, ensure that they are legible, do not obliterate the text on which you are commenting, and provide enough information to give the student a fair chance of doing better next time.

*Think of yourself as a teacher rather than an editor. Your goal is to support student learning and promote further thinking rather than marking every error. If work does need serious editing consider editing a paragraph or small portion as an example and explaining that this is what is required next time. This also works well if referencing is incorrect.

*Help students determine which of your comments are the most important. Students tend to apply equal weight to all comments, so you can help them by identifying the main areas for improvement and the aspects they have done well.

*Ensure your marking has been consistent over time. Have you marked the first assignments in much the same way as the last ones? If you are marking a large batch of assignments, you need to go back and ‘self-moderate’ to ensure that you are marking consistently.

*Take regular breaks to ensure that you stay fresh and focused. Reward yourself when you attain milestones such as ‘halfway there’.

III-Data analysis

In the analysis process, descriptive statistics are very useful for the teacher to intelligently summarize student results. Analyzing test results allows the teacher and all users to have access to summary data that can be easily interpreted instead of having long sheets of paper including student names and results. The first step in data analysis is to describe or summarize the data through descriptive statistics. For example, a teacher may be interested only in describing class performance and how well people are doing relative to average class performance. Descriptive statistics allow us to validly describe many results with few clues by giving answers to questions like: what was the average score on the test? were most of the ratings close to average or widely dispersed? And so on. Descriptive statistics also allow us to formulate interpretative statements (normative and criteria, but above all normative) on individual scores and to answer questions like: is this score above or below the average? how does this score deviate from the mean? what percentage of the scores was lower than this result? And so on. Whether locally designed or standardized instruments are administered, descriptive statistics are an important tool for interpreting the results of the results.

Measures of central tendency

A measure of central tendency is a summary statistic that represents the center point or typical value of a dataset. In statistics, the most common measures of central tendency are the median, mean, mode and range. Each of these measures calculates the location of the central point using a different method.

Example 1

Median (Middle) <i>The number which is in the middle or the middle value.</i> 11 7 11 18 9 7 6 23 7 6 7 7 7 9 11 11 18 23 Median: 9	Mode (Most) <i>The number that appears the most.</i> 11 7 11 18 9 7 6 23 7 6 7 7 7 9 11 11 18 23 Mode: 7
Mean (Average) <i>The total of the numbers divided by how many numbers there are.</i> 11 7 11 18 9 7 6 23 7 $11+7+11+18+9+7+6+23+7=99$ $99 / 9 = 11$ Mean: 11	Range (Difference) <i>The difference between the largest and the smallest number.</i> 11 7 11 18 9 7 6 23 7 Large : 23 Small : 6 $23 - 6 = 17$ Range: 17

Example 2

Mean, Median, Mode, & Range

~~5~~, 7, 7, 9, ~~11~~, ~~12~~, ~~14~~

Mean = $\frac{\text{Sum}}{n}$ Mode = 7

Range = $14 - 5$ Median = 9
 = 9

IV Interpretation of data

We have already explained that a result in itself is unimportant, unless it is analysed and interpreted using criteria. LoCascio points out three mistakes to avoid while interpreting evaluation results:

1- When two seemingly related events occur at the same time, it can be easy to assume that one event caused the other. For instance, a rise in student grades may coincide with the start of an afterschool homework club. Yet, what appears to be a logical relationship may actually be misleading information, and occasionally, when poorly interpreted, can convince schools to perpetuate ineffective practices and programs.

2-Failing to Understand the Intricacies of Averaged Data: The mean is created by adding together all the scores in a given set and dividing by the number of entries. In short, it is the sum divided by the count. Though easy to calculate and compare, the mean has a few drawbacks: it can be easily skewed by abnormally high or low scores as well as strongly influenced by the size of the count itself—averages based on very small groups have less validity than averages based on large groups. Due to these issues, the resulting average score can be misleading and, either in a positive or negative way, potentially create inaccurate descriptions of student performance.

3- Creating False Connections between Averaged Data and Real Students: Despite its connotation, there are actually very few students who are the living embodiment of averaged data. Most of our students have instructional needs—both strengths and weaknesses—which vary greatly from the average score. Yet both our informal opinions and our large scale school analysis tend to place a large emphasis on the averaged data. As a result we tend to create broad conclusions about student learning which do not necessarily reflect the strengths and needs of our real students.

V-Use of test results/making educative decisions

Assessment is the systematic collection and analysis of data for decision-making purposes. Thus, the results of the data analysis are used for decision making. The types of decisions that come after the evaluation are extremely important and varied and here are four examples: we can decide that a student X does not make satisfactory progress during this school stage and conclude that the education system of country Y does not reach her goals. We can equally conclude that student X is admitted to the next class or that teacher Y needs to change his teaching method. The following types of people, inside and outside the school system, are interested in the test results:

- a) Legislators
- b) School boards
- c) Managers
- d) Parents
- e) Students
- f) Teachers
- g) Decisions-makers

BIASES IN THE CORRECTION

When you correct, you make a judgement on the performance of your students (Durand and Chouinard, 2012). Different biases can affect this judgement. This part of the lesson summarizes the main biases, while offering suggestions to avoid them influencing you during the correction. Biases in correction can be related to various factors:

- the student
- individual copies
- the group
- the teacher

1-Factors related to the student

Elements	Description	Possible bias	Suggestion to avoid bias
Pygmalion effect (Knowledge of the student)	When the teacher knows the student, he can have a stereotypical image of him: good, lazy, limited, etc.	Over-evaluation or under-evaluation of the student's production according to the stereotype maintained.	When possible, the teacher makes sure to hide the student's name on the copy. Students can be asked to identify their work using a serial number.
The student's gender and appearance	Many studies in social psychology have shown the effect of an individual's appearance on the perception and conduct of others towards him. Even more evident if they are female.	Tall, thin and beautiful people clearly enjoy a favorable prejudice in society.	

2-Factors related to individual copies

Elements	Description	Possible bias	Suggestion to avoid bias
Tendency to approximation	Found especially in the case of correction of relatively long and complex written productions. Here, the teacher will tend to make global judgements which depart from the benchmarks.	Depending on whether the text generally appeals or not, the teacher will give it an approximate high or low score, without too much consideration of the criteria.	It is a question of determining the evaluation criteria well in advance, of defining well the grid, and of applying them rigorously. If there are general comments about production that are relevant to mention, they can always be added as annotations.
Halo effect	The teacher is quickly influenced by the first impression and, more precisely, by the external appearance of the production (its formal presentation).	Work with careful presentation will be more likely to be given a good rating and will decrease the severity of the correction in the event of unsatisfactory content.	The teacher requests the same form of presentation when the aesthetic character is not evaluated.
Persistence effect	The teacher will tend to evaluate all of a student's production based on the initial information he draws from it at the start.	For example, a student who makes some major mistakes in the first pages/lines of his work could see the teacher correct more severely and neglect his "successes" .	The teacher can make a first reading by examining a first evaluation criterion. Thus, at second reading, he dwells on another criterion and is less likely to be influenced by the first reading.

3-Factors related to the group

Elements	Description	Possible bias	Suggestion to avoid bias
Central tendency	Many teachers still more or less consciously maintain the idea that the results granted should correspond more or less to a distribution according to the normal curve.	They will be able to adjust their correction by being more severe if too many students obtain high results or by being more tolerant if too many students fail.	The teacher should stick to the criteria and make sure to rest when necessary.
Accumulation of corrected copies	It is not only a question here of the possibility that a teacher lets himself be influenced by the quality of the preceding copies, but also of the possibility that he lets himself influenced by the fact that it is the twentieth copy that he corrects without break.	Several possibilities of bias exist here since the order and the number of copies let in other biases such as fatigue. Thus, biases can range from overvaluation to undervaluation depending on the case.	

4-Factors related to the teacher

Elements	Description	Possible bias
Teacher's personality	If he is a demanding, rigorous, tolerant person, he may think that the student must respond to this model.	Certain laxity or excessive severity in the evaluation of production.
Teacher's physiological state	A teacher who is in good shape in the morning may not correct with the same glasses as the previous morning when he got up with headache. Same as fatigue, pain, energy, etc.	Could lead to sustained and insightful attention or on the contrary he can be negligent and failing in the evaluation of the student's copy.
Teacher's psychological and emotional state	The teacher's humor, emotional and psychological state have an effect on his attitude towards a student's work.	For example, an aggressive teacher in front of a "small" calligraphy may be more severe in a given week. If the same teacher receives good news the following week, he may be more tolerant of the same situation.

Conclusion

Finally, to avoid the combination of multiple factors, it is suggested to spread the correction, so as to arrive at realistic working sessions. Also, the teacher can afford to come back to a few corrected copies the day before in order to ensure the consistency of his correction.